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MujocoROS2Control: Seamless MuJoCo Integration with ROS 2 for Robot Simulation and Control

Adrian Danzglock [1], **Vamsi Krishna Origanti** [1], **and Frank Kirchner** [1,2]

1 Robotics Innovation Center, German Research Center for Artificial Intelligence (DFKI), Bremen, Germany **2** Universität Bremen, Bremen, Germany

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Software

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Summary

The `MujocoROS2Control` hardware interface enables seamless integration between MuJoCo (Todorov et al., 2012), a high-performance physics engine, and ROS 2 (ROS-Controls Team, 2024b), a widely adopted middleware for robotic systems. This interface provides an efficient solution for simulating and controlling robots using MuJoCo's physics capabilities within the ROS 2 ecosystem.

To support ROS-based workflows, we developed a dedicated URDF-to-MJCF conversion script. This tool translates URDF models into MJCF (MuJoCo XML format), preserving kinematic and dynamic properties and allowing custom MuJoCo-specific parameters such as sensors, actuators, and collision definitions to be specified directly in the URDF. This conversion ensures compatibility and adaptability for simulation.

`MujocoROS2Control` bridges the gap between ROS 2 and MuJoCo, offering a streamlined workflow for simulation, controller testing, and reinforcement learning.

Statement of Need

Developing and validating control algorithms for robotic systems often requires extensive testing, which on physical hardware can be expensive, time-consuming, and subject to wear. Accurate simulation environments are essential for safe and scalable development.

MuJoCo (Multi-Joint dynamics with Contact) is a fast and accurate physics engine designed for robotics, control, biomechanics, and reinforcement learning. It provides precise multi-body dynamics with advanced numerical integration, High-speed simulation for real-time control and machine learning, Sophisticated contact modeling and soft constraint handling, Flexible actuator models and comprehensive sensor support.

MuJoCo is ideal for reliable torque feedback due to its high-fidelity torque-controlled joints and smooth-contact soft constraint solver, which enables precise compliance behavior (Zhang et al., 2025).

Gazebo, on the other hand, typically relies on hard-constraint engines like ODE or Bullet, which struggle with compliant torque-based tasks unless carefully tuned. These engines often require smaller timesteps than MuJoCo to maintain stability (Team, 2025).

Drake supports torque control, but due to its symbolic and rigid-body emphasis, tuning compliance can be computationally intensive (Tedrake & Drake Development Team, 2019).

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Table 1: Comparison of usability of different Simulators for usage with Compliant controllers

Controller Type	**MuJoCo**	**Gazebo/Ignition**	**Drake**
Torque-based impedance	High-fidelity, fast, stable	Accurate but tuning required	Accurate, symbolic, slower than mujoco
Admittance with wrench input	Stable, smooth compliant	Plugin-lag, tuning required	precise, heavier computationally
Time-step flexibility	Larger stable steps (e.g. 2-5ms)	Smaller steps required (1ms)	Variable but slower dynamics

Table 2: Comparison of actual ros2 simulator wrappers

Feature	**Mujoco ROS2 Control**	**mujoco ros2 control (MoveIt, 2025)**	**gz ros2 control (ROS-Controls Team, 2025)**	**drake-ros (Locomotion, 2025)**
Simulation Engine	MuJoCo	MuJoCo	Gazebo	Drake
Sensor Support	Yes, IMU, Pose, Wrench, RGBD	No, Planned	Yes, Supports various sensors via Gazebo plugins	Yes, but aditional code is required
URDF Support	Yes, direct loading from URDF via urdf to mjcf script in launchfile	No, Planned	Yes, Uses URDF/SDF for robot models	Yes, Supports URDF and custom formats
Control System	ros2_control	ros2_control	ros2_control	Experimental API for ROS2
Control Methods	PID, Mujoco Actuators, Torque	PID, Torque, not integrated position and velocity control	PID, Effort, Position, Velocity	PID, Optimazation-based-control
Mimic Joints	Yes	Yes	Yes, sometimes difficoult to setup (User, 2025)	Yes

Despite these capabilities, MuJoCo lacks native support for ROS 2, limiting its adoption in modern robotic development pipelines. `MujocoROS2Control` addresses this gap, enabling users to simulate ROS 2-compatible robots in MuJoCo with minimal overhead.

Implementation

`MujocoROS2Control` integrates MuJoCo with the `ros2_control` framework. A key component is the URDF-to-MJCF converter, which maintains fixed joints which MuJoCo typically collapses, allows sensor and actuator tags within URDFs, and generates MJCF files compatible with MuJoCo's expectations.

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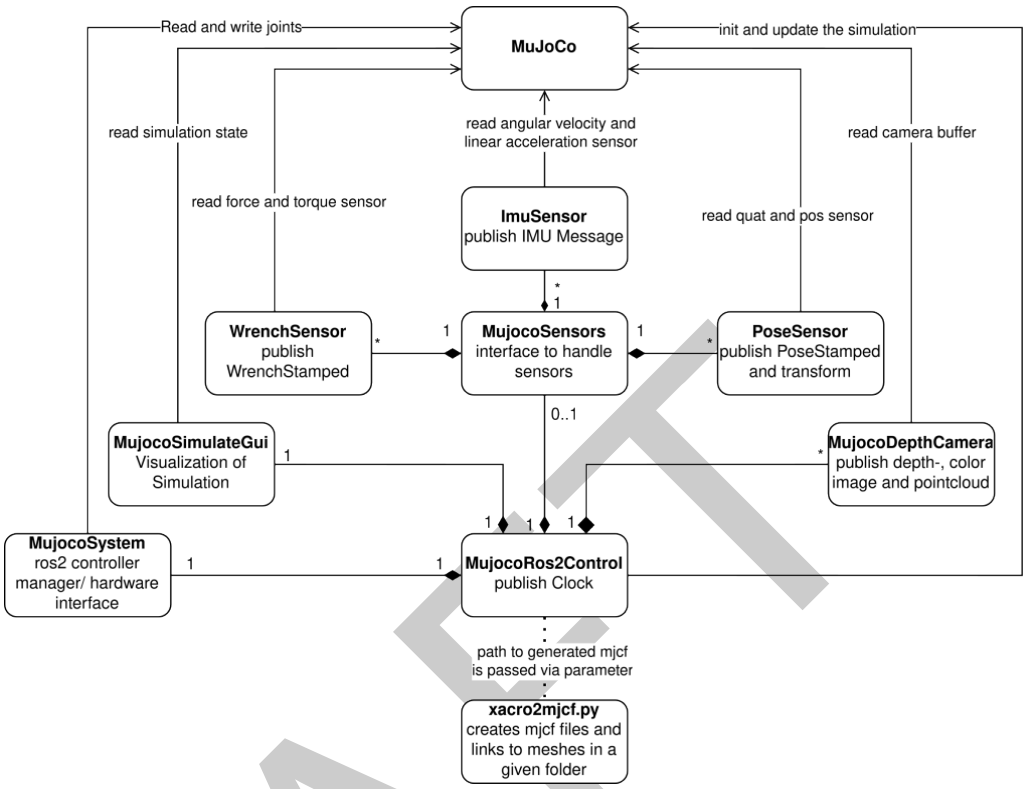

Figure 1: Overview of the MujocoRos2Control Structure

The interface supports direct torque control, PID-based position/velocity/acceleration control and MuJoCo's native actuator models.

Joint states and simulation time are published for synchronization with the ROS 2 system time (`/clock`). Sensors defined in the URDF are exposed as individual ROS nodes using `realtime_tools` (ROS-Controls Team, 2024a) to maintain real-time performance.

Use Cases

MujocoRos2Control was utilized for testing and validating various torque and admittance controllers within the scope of the HARTU project (*HARTU Project*, n.d.). The software also played a key role in conducting experiments for the publication "Look-Ahead Optimization for Managing Nullspace in Cartesian Impedance Control of Dual-Arm Robots" (Origanti et al., 2025).

Our framework has been successfully employed to test components such as force-torque sensor gravity compensation, torque-based Cartesian controllers, and Dynamic Movement Primitives (DMP)-based skill reproduction (Fabisch, 2024).

Examples

Franka FR3 with IndustRealKit Gears

This example integrates the Franka FR3 robot (FRANKA ROBOTICS, 2024) with high-resolution gear models from the IndustRealKit (Tang et al., 2023). MuJoCo actuators are used to generate joint torques, although the implemented PID control and torque control are also supported. For the high-resolution collision modeling, we use CoaCD (Wei et al., 2022) to

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create multiple convex hulls from complex mesh geometry. The URDF-to-MJCF converter automatically replaces original mesh files in the mjcf, when the converted files are in the same directory.

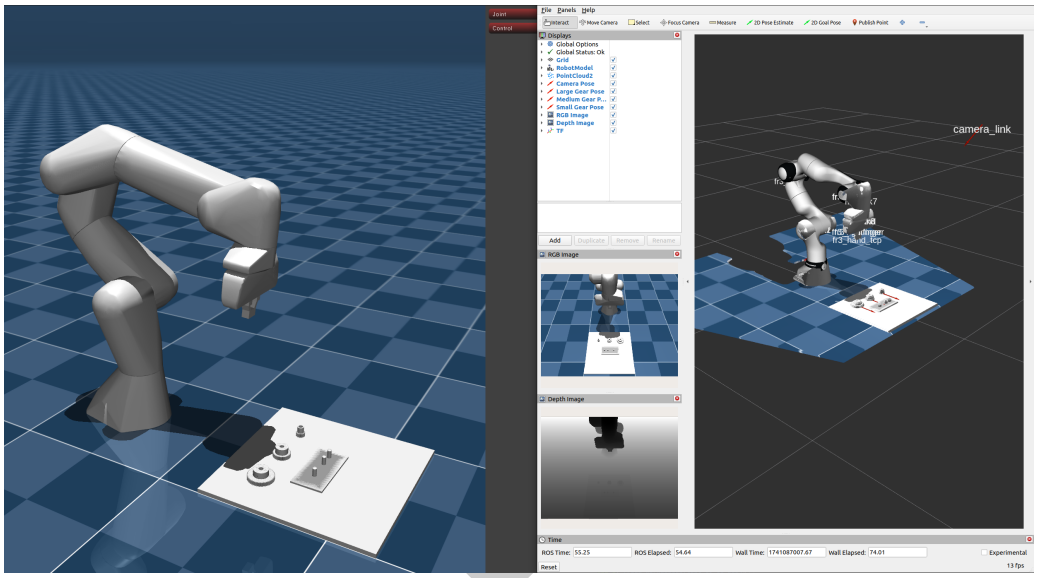

Figure 2: Franka FR3 controlled with ROS 2 Joint Trajectory Controller

Unitree H1

This example uses the loads the Bipedal robot Unitree H1 (Robotics, 2025) with a floating base and tf2 transformations from world to pelvis. All joints are controlled via ros2 control, with mujoco actuators for position and velocity control (PD Control).

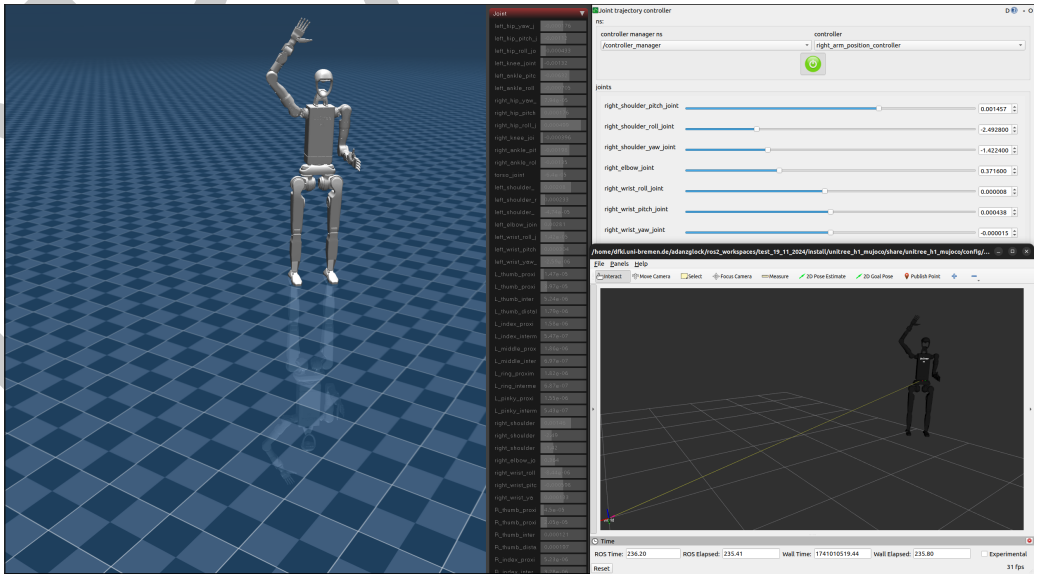

Figure 3: Unitree H1 controlled with ROS 2 control and tf2 transformation, and IMU

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IMRK System

In this example, we simulate the iMRK system developed at DFKI Bremen, consisting of two KUKA LBR iiwa 14 robots (Dennis Mronga, 2025). A ROS2 Cartesian impedance controller (migrated version of (Mayr & Salt-Ducaju, 2024)) is used for each arm. MuJoCo actuators manage the Robotiq 2F grippers, and force-torque sensors are simulated at both end-effectors. Because the robot description for the IMRK system is not public available, we cant provide the files for this example.

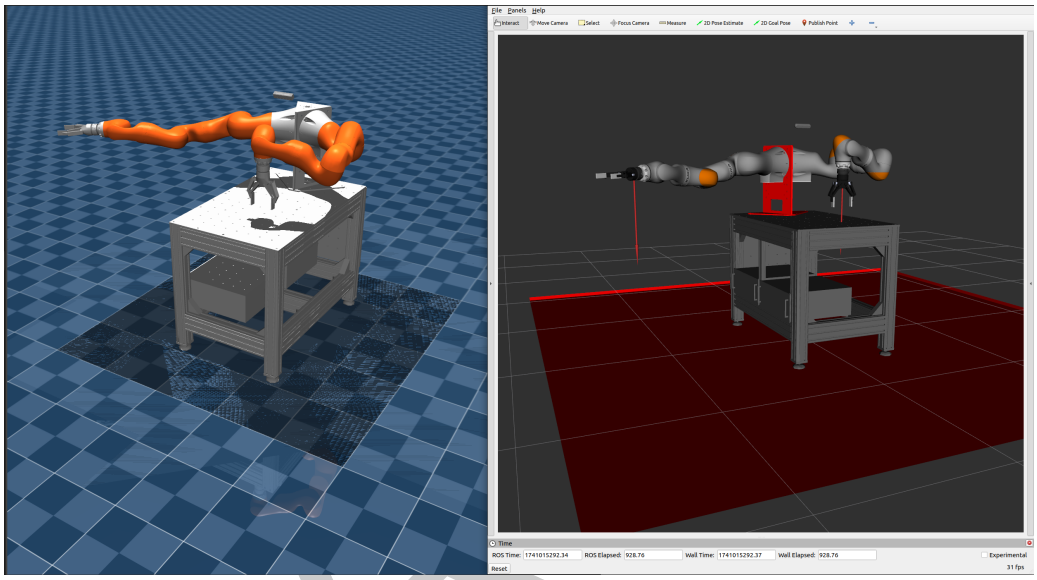

Figure 4: IMRK with Robotiq 2F Gripper and Robotiq FT300 Sensor

Conclusion

`MujocoROS2Control` enables high-fidelity robotic simulation by integrating MuJoCo with ROS 2. Through its robust conversion utilities, sensor bridging, and actuator support, it provides a reliable framework for control development, validation, and machine learning research.

Its lightweight design and fast simulation performance make it well-suited for force-based control, trajectory optimization, and reinforcement learning applications. Future enhancements may include support for additional sensor types, multi-robot coordination, and real-time closed-loop learning.

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